Library Management System

Aim:

To analyze the Software Development Life Cycle (SDLC) models used in the development of a Library Management System and determine the most suitable SDLC model for its successful implementation.

Objective:

- Study different SDLC models.
- Analyze the development process of a Library Management System.
- Compare the advantages and limitations of various SDLC models.
- Recommend the most appropriate SDLC model for the application.

Selected Software Application:

(Library Management System)

Overview

- A Library Management System is a software application designed to automate library operations and improve efficiency for librarians and users. It enables users to search books, borrow and return books, reserve books online, pay overdue fines, and manage memberships. The system also helps librarians maintain book inventories, issue books, generate reports, and monitor library activities.

Functional Requirements :

Student/User Module;

- User Registration
- Secure Login
- Search Books
- View Book Availability
- Borrow Books

Librarian Module:

- Add New Books
- Update Book Details

- Delete Books

Admin Module:

- User Management
- Staff Management
- Library Reports

Why Agile is Best?

Reasons

- Library requirements change frequently.
- New features can be added regularly.
- Bugs can be fixed quickly.
- Digital libraries require continuous updates.

SDLC Phases:

1. **Requirement Analysis**
   - Requirements Collected
   - Student Requirements
   - Librarian Requirements
   - Book Database

2. **System Design**

- Design Includes
- Database Design
- User Interface Design
- System Architecture

Conclusion:

The case study concludes that selecting the appropriate SDLC model depends on the project requirements. Since a Library Management System requires frequent updates, user feedback, security improvements, and continuous maintenance, the Agile SDLC model is the most suitable choice. When combined with DevOps practices, it provides faster deployment, improved software quality, reduced development risks, and efficient maintenance. This combination ensures that the Library Management System remains reliable, secure, scalable, and capable of meeting the evolving needs of students, librarians, and administrators.